

Professional Implementation Guide

AI Voice Booking System for Car Detailing and Cleaning Businesses

A 4-week, step-by-step execution guide for a recently graduated computer engineer. This document explains what to build, in what order, the expected deliverables each week, and the best official and reputable learning resources for each task.

Project Goal Build a multilingual voice-booking system that handles inbound calls, IVR/keypad menus, quote generation, availability checks, booking creation, cancellations, reminders, provider summaries, and basic admin operations.	Recommended Stack Twilio Voice + TwiML/Gather, n8n, PostgreSQL, Google Calendar API, Next.js dashboard, and AI only where useful (language handling, transcript cleanup, fallback responses).
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Important rule for the engineer: keep pricing, slot generation, booking creation, cancellation, and reminder timing deterministic. Use AI only where it improves usability, not for core business logic.

What success looks like

- Customer can call, choose a language, choose a service, hear a quote range, hear available slots, and confirm a booking.
- Customer can cancel an upcoming booking and the slot becomes available again.
- Customer receives confirmation and day-before reminders.
- Service provider receives booking confirmation, day-before reminders, and a daily 4:30 PM summary for tomorrow's schedule.
- The same template can be configured for either car detailing or cleaning businesses.

Recommended learning sources to trust first

Topic	Primary website	YouTube / channel
Twilio Voice / TwiML / Gather	Official docs https://www.twilio.com/docs/voice/twiml	TwilioDevs channel https://www.youtube.com/twiliodevs
n8n webhooks and workflow automation	Official docs https://docs.n8n.io/integrations/builtin/core-nodes/n8n-nodes-base.webhook/	Official n8n channel https://www.youtube.com/c/n8n-io
Google Calendar API	Official docs https://developers.google.com/workspace/calendar/api/v3/reference	Google for Developers resources https://www.youtube.com/playlist?list=PLw5h0DiJ-9PCKQ3LYSnGOGjV-HLngegup
Ollama local API	Official docs https://docs.ollama.com/api/introduction	Official docs preferred over video https://docs.ollama.com/api/introduction

Topic	Primary website	YouTube / channel
PostgreSQL fundamentals	Official docs https://www.postgresql.org/docs/current/tutorial.html	PostgresTV https://www.youtube.com/PostgresTV
Next.js / dashboard UI	Official docs https://nextjs.org/docs/app/getting-started	Vercel channel https://www.youtube.com/@VercelHQ

Week 1 — Foundation, environment setup, schema, and architecture

Week goal: Stand up the technical foundation: repository, runtime, database, business configuration model, Twilio basics, n8n instance, and a working local development environment.

Create project repository and environment

- Create a Git repository with folders for /backend, /workflows, /dashboard, /docs, and /sql.
- Prepare a .env template for Twilio, Google, database, email, and AI configuration values.
- Write a short architecture note describing the final stack and the split between deterministic logic and AI-assisted logic.

Learn this task properly:

- [Next.js getting started](#)

<https://nextjs.org/docs/app/getting-started>

- [n8n docs home](#)

<https://docs.n8n.io/>

- [PostgreSQL getting started](#)

<https://www.postgresql.org/docs/current/tutorial-start.html>

Set up PostgreSQL and core schema

- Create the main tables: businesses, services, customers, bookings, calls, booking_events, availability_rules, notification_log, callback_requests.
- Add indexes for twilio_call_sid, caller phone, booking status, and booking date/time.
- Seed one demo business for car detailing and one for cleaning.

Learn this task properly:

- [PostgreSQL tutorial](#)

<https://www.postgresql.org/docs/current/tutorial.html>

- [PostgresTV channel](#)

<https://www.youtube.com/PostgresTV>

Set up n8n and webhook strategy

- Run n8n locally or on a test server.
- Create a hello-world webhook workflow that receives POST data and responds successfully.
- Document test URL vs production URL handling so Twilio later uses the production endpoint.

Learn this task properly:

- [Webhook node docs](#)

<https://docs.n8n.io/integrations/builtin/core-nodes/n8n-nodes-base.webhook/>

- [Respond to Webhook docs](#)

<https://docs.n8n.io/integrations/builtin/core-nodes/n8n-nodes-base.respondtwebhook/>

- [Official n8n channel](#)

<https://www.youtube.com/c/n8n-io>

Set up Twilio and verify inbound call plumbing

- Buy or configure a Twilio voice number.
- Point the number's voice webhook to the n8n production URL.
- Test that a basic TwiML response plays on a real phone call.

Learn this task properly:

- [Twilio Voice / TwiML docs](#)

<https://www.twilio.com/docs/voice/twiml>

- [Twilio Gather docs](#)

<https://www.twilio.com/docs/voice/twiml/gather>

- [TwilioDevs channel](#)

<https://www.youtube.com/twiliodevs>

Define reusable business configuration

- Create a JSON or DB-backed config per business: languages, main menu, service catalog, durations, quote rules, reminder timing, provider email, calendar id, service area, slot size.
- Create one sample config for car detailing and one for cleaning.

Learn this task properly:

- [Google Calendar API reference](#)

<https://developers.google.com/workspace/calendar/api/v3/reference>

- [Next.js docs](#)

<https://nextjs.org/docs>

Expected deliverables this week

- Git repository initialized with project structure.
- PostgreSQL schema created and seeded with sample businesses and services.
- n8n reachable through a test and production webhook.
- Twilio number connected to n8n and returning a simple voice response.
- A written architecture note and business configuration template.

Week 2 — Quote engine, availability engine, and core service logic

Week goal: Build the deterministic business logic that powers quotes, slot generation, and slot suggestion. This week should produce working backend services before the full voice flow is attempted.

Build quote engine for car detailing and cleaning

- Create rules for car detailing using service type, vehicle type, condition, add-ons, and suburb/travel fee.
- Create rules for cleaning using service type, property type, bedrooms, bathrooms, extras, suburb/travel fee, and frequency.
- Return quote_min, quote_max, and a manual_review flag.

Learn this task properly:

- [PostgreSQL tutorial](#)

<https://www.postgresql.org/docs/current/tutorial.html>

- [n8n docs](#)

<https://docs.n8n.io/>

Connect Google Calendar and build availability engine

- Create OAuth credentials and connect Google Calendar.
- Use freeBusy.query to read provider busy blocks.
- Combine calendar data with availability rules and service duration to generate the best three slots.

Learn this task properly:

- [Google Calendar freeBusy query](#)
<https://developers.google.com/workspace/calendar/api/v3/reference/freebusy/query>
- [Google Calendar create events guide](#)
<https://developers.google.com/workspace/calendar/api/guides/create-events>
- [Google for Developers playlist](#)
<https://www.youtube.com/playlist?list=PLw5h0DiJ-9PCKQ3LYSnGOGjV-HLnnyegup>

Create test workflows in n8n

- Build one workflow that takes structured request data and returns a quote.
- Build a second workflow that takes service duration and desired date window and returns three suggested slots.
- Store test results in PostgreSQL for traceability.

Learn this task properly:

- [n8n webhook docs](#)
<https://docs.n8n.io/integrations/builtin/core-nodes/n8n-nodes-base.webhook/>
- [n8n YouTube channel](#)
<https://www.youtube.com/c/n8n-io>

Add local AI service for non-critical tasks

- Install Ollama and test local API access.
- Create a small helper service that can normalize speech or free-text into structured fields.
- Do not allow AI to directly write bookings or prices.

Learn this task properly:

- [Ollama API intro](#)
<https://docs.ollama.com/api/introduction>
- [Ollama local auth behavior](#)
<https://docs.ollama.com/api/authentication>

Expected deliverables this week

- Working quote engine for both business types.
- Working availability engine returning at least three slots.
- n8n workflows that can be tested independently with structured JSON.
- Ollama installed and callable for helper tasks.

Week 3 — Twilio call state machine, booking flow, and cancellation flow

Week goal: Turn the backend logic into a working phone system with multilingual menus, stateful routing, booking confirmation, and cancellations.

Build the Twilio route table and state machine

- Create states such as LANGUAGE_MENU, MAIN_MENU, SERVICE_MENU, COLLECT_REQUIRED_FIELDS, QUOTE_RESULT, CHECK_AVAILABILITY, CONFIRM_SLOT, CREATE_BOOKING, IDENTIFY_BOOKING_CANCEL, CANCEL_CONFIRM, CANCEL_BOOKING, CALLBACK_REQUEST, END_CALL.
- Store current_state in the calls table and update it after every valid caller action.
- Support both DTMF Digits and SpeechResult parsing.

Learn this task properly:

- [Twilio Gather docs](#)

<https://www.twilio.com/docs/voice/twiml/gather>

- [Twilio TwiML overview](#)

<https://www.twilio.com/docs/voice/twiml>

- [TwilioDevs channel](#)

<https://www.youtube.com/twiliodevs>

Build multilingual menu prompts

- Add English first, then Hindi, then Mandarin prompts for language menu and main menu.
- Keep prompts short and phone-friendly.
- Ensure 9 repeats the current menu.

Learn this task properly:

- [Twilio TwiML docs](#)

<https://www.twilio.com/docs/voice/twiml>

- [ConversationRelay docs for future enhancement](#)

<https://www.twilio.com/docs/voice/conversationrelay>

Create booking workflow

- After quote and slot selection, recheck slot availability.
- Create booking record in PostgreSQL.
- Insert Google Calendar event and store its event id.
- Send confirmation to customer and provider.

Learn this task properly:

- [Google Calendar events.insert](#)

<https://developers.google.com/workspace/calendar/api/v3/reference/events/insert>

- [Google Calendar create events guide](#)

<https://developers.google.com/workspace/calendar/api/guides/create-events>

- [n8n docs](#)

<https://docs.n8n.io/>

Create cancellation workflow

- Find future bookings by caller phone number.
- Read booking details back to the customer.
- Require explicit confirmation before cancellation.
- Cancel the booking, update Google Calendar, and send notifications.

Learn this task properly:

- [Google Calendar API reference](#)

<https://developers.google.com/workspace/calendar/api/v3/reference>

- [Twilio Gather keypad tutorial](#)

<https://www.twilio.com/docs/voice/tutorials/how-to-gather-user-input-via-keypad>

Create callback request fallback

- If the customer is confused or asks for help, collect a spoken callback reason and save it to `callback_requests`.
- Notify the provider by email.

Learn this task properly:

- [n8n docs](#)

<https://docs.n8n.io/>

- [Twilio docs](#)

<https://www.twilio.com/docs>

Expected deliverables this week

- A working voice flow from call start to quote to slot selection to booking creation.
- A working cancellation flow.
- English menus fully working; Hindi and Mandarin menus at least added for language/main menu.
- Callback request fallback saved in the database.

Week 4 — Reminders, provider summary, dashboard, testing, and handover

Week goal: Make the system operational for real business use: reminders, provider summary at 4:30 PM, basic dashboard, end-to-end tests, bug fixes, and handover documentation.

Build scheduled reminders and daily provider summary

- Create a customer reminder workflow that runs about 24 hours before the booking.
- Create a provider reminder workflow about 24 hours before the booking.
- Create a provider summary workflow that runs at 4:30 PM Melbourne time and lists tomorrow's bookings and totals.

Learn this task properly:

- [n8n schedule trigger docs](#)

<https://docs.n8n.io/integrations/builtin/core-nodes/n8n-nodes-base.scheduletrigger/>

- [Google Calendar API reference](#)

<https://developers.google.com/workspace/calendar/api/v3/reference>

Build minimal admin dashboard

- Create a Next.js dashboard with pages for Dashboard, Bookings, Tomorrow, Calls, Callbacks, Services, Pricing, Availability, and Settings.
- Read data from PostgreSQL and make it easy for the business owner to verify bookings and callbacks.
- Do not overbuild styling; prioritize clarity and working filters.

Learn this task properly:

- [Next.js docs](#)

<https://nextjs.org/docs>

- [Next.js learn](#)

<https://nextjs.org/learn>

- [Vercel YouTube channel](#)

<https://www.youtube.com/@VercelHQ>

Test all major scenarios professionally

- Write a test matrix covering: new booking, quote only, availability only, booking confirmation, cancellation, callback, invalid keypad entry, speech fallback, no response, and reminder scheduling.
- Test each scenario for both business types.
- Capture defects and fix them before sign-off.

Learn this task properly:

- [Twilio docs](#)

<https://www.twilio.com/docs>

- [n8n docs](#)

<https://docs.n8n.io/>

- [PostgreSQL docs](#)

<https://www.postgresql.org/docs/>

Prepare handover package

- Document environment variables, service accounts, OAuth steps, Twilio number setup, n8n workflow names, database migration files, and the exact deployment process.
- Create an operator guide for the business owner.
- Create a developer handover note explaining where to modify services, prices, prompt text, and reminder timing.

Learn this task properly:

- [n8n webhook reverse proxy notes](#)

<https://docs.n8n.io/hosting/configuration/configuration-examples/webhook-url/>

- [Next.js getting started installation](#)

<https://nextjs.org/docs/app/getting-started/installation>

Expected deliverables this week

- Customer and provider day-before reminders working.
- Provider 4:30 PM tomorrow-summary email working.
- Minimal provider dashboard working.
- Full end-to-end test checklist completed and defects fixed.
- Deployment + handover documentation ready.

Professional execution checklist

- Keep all core workflows versioned in Git and export n8n workflows after every stable milestone.
- Use database migrations instead of manual table edits.
- Create sample test data for both a car-detailing business and a cleaning business.
- Record every external dependency: Twilio number, Google Calendar credentials, email provider, database connection, and domain/webhook URL.
- Document every prompt, business rule, and state transition in plain English.
- Test with real phones, not only simulated requests.
- Use a defect log and close all P1/P2 issues before handover.

Minimum handover package

Item	What must be included
Source repository	Backend code, SQL migrations, dashboard code, README, environment template.
n8n exports	All workflows exported with clear names and the order they should be imported.
Database setup	Schema, seed data, migration instructions, and backup/restore notes.
Twilio setup guide	Phone number setup, webhook path, test procedure, and production URL.
Google setup guide	OAuth credentials, calendar permissions, freeBusy and event creation setup.
Operations guide	How the business owner views bookings, callbacks, tomorrow's jobs, and reminders.

Final advice to the graduate engineer

Do not start by making the voice agent sound impressive. Start by making the booking system correct. A reliable but simple caller experience is more valuable than a human-like but error-prone one. Build deterministic business logic first, then add multilingual polish, then add AI fallback only where it clearly improves the user experience.